CROP STRESS PHYSIOLOGIST · RESEARCH PORTFOLIO

RANADHEER REDDY VENNAM, Ph.D.

Research Assistant, Tidewater Agricultural Research and Extension Center, Virginia Tech · Ph.D. Plant Pathology, Physiology and Weed Science (2026)

A season's yield is decided in a few weeks around flowering. I study what **heat and drought** do in that window, what they cost growers, and how to see the damage from the air.

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$680–770	52–73%	R^2 0.89	~3,190	14
per acre lost to heat + drought	pod yield lost to a hot, dry mid-season	drone wilting prediction, unseen season	drone-imaged plots, 12 environments	peer-reviewed papers, 8 first-author
peanut · 2024 & 2025	rainout shelters · Tidewater AREC	spectral bands only	3 locations · VA & NC	incl. 1 in press · 2 in revision

RESEARCH: FOUR QUESTIONS, ASKED IN SEQUENCE



01 · When does water stress do the most damage to corn?

MS · Mississippi State · 2021–2023 · co-designed, ran, analyzed

Stage-resolved drought and salinity response curves: 14-day deficits at flowering or grain fill across five soil-moisture levels, a seven-day drought with rewatering, and early-season water and salinity dose responses.

–53%	kernel number, 14 days of drought at flowering
–46%	chlorophyll after 7 days; did not recover
3 dS/m	early-season maize salinity threshold



02 · Is heat plus drought just the sum of its parts?

MS collaborations + Ph.D. · designed and led the peanut chamber-vs-field study

Combined stress in maize, soybean and peanut. Paired growth-chamber and rainout-shelter experiments on five Virginia-type peanut cultivars tested whether chamber screens hold up in the field.

–93%	soybean stomatal conductance, heat + drought
65 vs 21%	peanut conductance drop, chamber vs field
NC 20	and Sullivan ranked top in both settings



03 · Can a drone score stress in a season it has never seen?

Ph.D. · Virginia Tech · 2023–2026 · designed and led · FAA Part 107

UAV multispectral imagery at 3 locations in Virginia and North Carolina over 3 seasons, paired with 9,387 plot-level wilting scores; four ML algorithms tested with leave-one-year-out validation.

R^2 0.89	wilting predicted in a withheld season
12	environments, ~3,190 plots
8	environments inside the regional variety-testing program



04 · What does a hot, dry mid-season cost a peanut grower?

Ph.D. · Tidewater AREC · 2024–2025 · led · Peanut Science (in press), VCE SPES-672

Two seasons, 12 genotypes, movable rainout shelters for 65–70 days from flowering to early seed fill; yield, grade, gross value and net returns under four production-cost scenarios.

–52/–73%	pod yield, 2024 / 2025
$770	gross value lost per acre (2024)
8.2 t/ha	top rainfed line, 14x009-1-5-1-1

SELECTED PUBLICATIONS (FIRST AUTHOR UNLESS NOTED)

1. Vennam & Balota (2026). Yield, grade, and economic performance of peanut under combined heat and drought stress. *Peanut Science*, in press. LED
2. Vennam et al. (2026). Leaf wilting as a phenotypic indicator of heat and drought stress in crops. *Discover Agriculture* 4:32. WROTE
3. Vennam et al. (2025). Peanut physiological responses to heat and drought across growth chamber and field environments. *Plants* 14:2687. DESIGNED, LED
4. Vennam et al. (2025). Drought and salt stress effects on physiology, leaf reflectance, and growth of *Brassica carinata*. *Industrial Crops and Products* 235. ANALYZED, WROTE
5. Vennam et al. (2024). Early-season maize responses to salt stress. *J. Agriculture and Food Research* 15. RAN, ANALYZED
6. Vennam et al. (2023). Soil moisture stress during silk emergence and grain filling in maize. *Physiologia Plantarum* 175. CO-DESIGNED, RAN
7. Vennam et al. (2023). Functional relationships between soil moisture and corn early-season growth. *Plants* 12:2471. RAN, ANALYZED
8. Poudel, Vennam et al. (2023). Resilience of soybean cultivars to drought during flowering. *Scientific Reports* 13:1277. CO-AUTHOR: DATA COLLECTION
9. Bheemanahalli, Vennam et al. (2022). Post-flowering heat and drought in maize (*Zea mays* L.). *Plant Stress* 6. CO-AUTHOR: ANALYZED, CO-WROTE
10. In revision: *Smart Agricultural Technology* (drone ML wilting) and *J. Sustainable Agriculture and Environment* (peanut traits and yield). DESIGNED, LED

RESEARCH VISION

Predict stress before it costs yield

Season-independent drone models plus weather and soil data for in-season warnings.

Breed for the combination

Screen advanced lines under combined heat and drought, before stress erases differences.

Connect mechanism to phenotype

Link leaf gene expression to wilting, gas exchange, and reflectance.

RECOGNITION

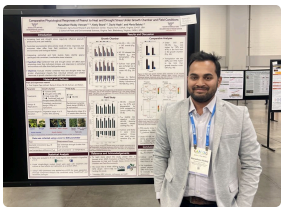
2026	Gerald O. Mott Award, Crop Science Society of America
2025	Outstanding Ph.D. Student, Association of Agricultural Scientists of Indian Origin
2025	Selected mentee, Bayer Crop Science B4U Mentoring Program
2024	Outstanding Alumni Award for Excellence in Research, Lovely Professional University
2022–26	Ten poster awards at national and regional meetings, four first place

EXTENSION, MENTORING, SERVICE

2025	VCE SPES-672: heat and drought costs for Virginia peanut growers
Field	On-farm trials with growers in VA and NC; Tidewater AREC field days
2026	Drone sessions for ~110 Suffolk Public Schools students
Lead	President, APRES Graduate Student Organization (2025–26)
Mentor	Undergraduate mentee co-authored a corn drought paper; supervised summer field workers
2026	Session judge, CANVAS 2026, Portland, OR: Precision Agriculture Community; Airborne and Satellite Remote Sensing
Review	29 manuscripts for 13 journals

EDUCATION

2026	Ph.D., Plant Pathology, Physiology and Weed Science, Virginia Tech (4.0)
2023	M.S., Agronomy, Mississippi State University (4.0)
2019	B.Sc. (Hons.) Agriculture, Lovely Professional University




Interactive version with simulator, gallery and all publications: see the portfolio website. Figures as of November 2026.

Ranadheer Reddy Vennam, Ph.D.